Session 3: AI Meets Your Data

Business Intelligence & AI

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From Personal Productivity to Enterprise Data

The Shift

You have spent a week using AI for **personal productivity** — charts, documents, workflows. Now we scale up. The same AI that drafts your memos can also **query your company's databases** and **search your company's documents**.

> But enterprise data is scattered across 5, 10, sometimes 20 systems — and trapped in policies, contracts, and board minutes nobody can search. Today we see what that looks like and what AI can do about it.

The 3-Week Email Chain

“Which of Our Customers Buy from Multiple Divisions?”

The CEO’s Question

- Simple enough to ask in one sentence
- Requires data from 3 separate CRMs
- Nobody owns the cross-system view
- The answer should drive pricing strategy

The 3-Week Reality

- Week 1: BI team extracts from Salesforce
- Week 2: Energy team pulls from legacy CRM
- Week 3: Manual matching in Excel, finally delivered
- **CEO has moved on to a different priority**

Demo: Dashboard vs. Chat

Introducing Meridian Corp

The Simulation Environment

- **$500M B2B industrial distributor** with 3 operating divisions
- **10 enterprise systems**: Salesforce, Legacy CRM, HubSpot, NetSuite, SAP, Oracle SCM, Workday, Zendesk, QuickBooks
- **41 tables, 26,000+ rows, 3 years of data**
- Realistic cross-system friction: different naming conventions, date formats, column styles
- **Plus a document collection**: employee handbook, vendor contracts, board minutes, strategic plans

You will log in and query this environment yourself in 15 minutes. First, watch what happens when we ask it the CEO’s question.

Live Demo: “Which Customers Buy from Multiple Divisions?”

The prompt

```
Which of our customers buy from multiple divisions?
Show combined revenue and flag any name mismatches across systems.
```

Demo Result: Cross-System Merge

Customer	Industrial (SF)	Energy (Legacy)	Safety (HubSpot)	Combined Revenue	Divisions
General Electric	General Electric	GE Industrial Solutions	GE Safety Division	$18.4M	3
ExxonMobil	ExxonMobil Corp	Exxon Mobil Energy Svcs	ExxonMobil Safety	$14.7M	3
Dow Chemical	Dow Chemical Co.	Dow Energy Products	Dow Inc.	$12.1M	3
Chevron	Chevron USA Inc.	Chevron Energy	Chevron Phillips Safety	$11.3M	3
3M Company	3M Industrial	—	3M Safety Products	$8.6M	2

10 customers buy from 2+ divisions. **4** buy from all 3 — representing **$56.5M** (11.4% of total revenue). Only 3 of 10 had matching names across systems.

Behind the Scenes

1

Query SF

Pull customers + revenue from Salesforce CRM

→

2

Query Legacy

Pull from Energy division's custom CRM

→

3

Query HubSpot

Pull from Safety division's CRM

→

4

Fuzzy Merge

Python matches names across systems (GE = General Electric)

The agent **cannot** join across systems in a single query. It pulls from each, then uses Python to reconcile. This mirrors how a real enterprise AI tool works.

The Business Lens

Remember the SaaSpocalypse?

What the Market Saw

- AI agents querying enterprise data in plain English
- Instant charts and narratives from a chat interface
- Cross-system answers that used to require BI teams
- **$2 trillion in software market cap erased**

What You Just Saw

- The Meridian Corp app answered in 30 seconds what took 3 weeks
- Fuzzy matching across 3 CRMs with different naming conventions
- Combined revenue view that no single dashboard could produce
- **This is the technology behind the valuation collapse**

The Implication

Dashboards

- Pre-built views of pre-selected metrics
- Cannot answer questions they were not designed for
- Every new question requires a development cycle

BI Teams

- Skilled analysts writing SQL and building reports
- Bottleneck: 3-week turnaround for cross-system questions
- High value, but limited bandwidth

AI Agents

- Any question, any time, in plain English
- Cross-system queries with automatic reconciliation
- **Not a replacement — a force multiplier**

Hands-On: Your First Enterprise Queries

Getting Started

Log In Now

1. Open your browser and navigate to the Meridian Corp app URL (in chat)
2. Log in with your provided credentials
3. You should see a chat interface — type your first question

Start with single-system queries to get comfortable. Then we will push into cross-system territory.

Round 1: Single-System Queries

Salesforce (Industrial CRM)

- “What is the total pipeline value for the Industrial division?”
- “Who are the top 5 customers by revenue in 2025?”

NetSuite (Finance)

- “What was total revenue by quarter for 2025?”
- “Show operating expenses by category for Q4”

Workday (HR)

- “How many employees does each division have?”
- “What is the average tenure by department?”

Round 2: Cross-System Queries

Try These

- “Which customers buy from multiple divisions?”
- “What is revenue per employee by division?”
- “Compare support ticket volume to revenue by division”
- “Which division’s headcount grew fastest relative to revenue?”

Watch For

- The agent querying multiple systems sequentially
- Python merge steps appearing in the response
- Fuzzy name matching across CRMs
- Narrative that synthesizes data from different sources

Demo Result: Revenue per Employee

Division	2025 Revenue	Headcount	Revenue / Employee
Energy	$52.1M	134	**$388,800**
Industrial	$87.3M	218	**$400,500**
Safety	$34.8M	87	**$400,000**
Company-wide	**$174.2M**	**482**	**$361,400**

Company-wide average is **10% below** divisional averages because 43 Corporate HQ employees generate no direct revenue. All divisions cluster tightly around $390K-$401K per employee.

Timed Challenge

5 Minutes

> “Which division’s top customer represents the greatest concentration risk — and what does their support ticket history look like?”

This requires CRM data (customer revenue) + Zendesk data (support tickets). Go.

What We're Skipping

The 60-90 Days the Demo Didn’t Show

Data Cataloging

- Inventory every table, column, and relationship
- Document what each field actually means
- Map data quality issues

Schema Documentation

- Write descriptions the LLM can understand
- The system prompt is “institutional knowledge”
- Gets it wrong? Fix the schema description

API Access

- Provision read-only database connections
- Set up authentication and rate limits
- Security review for each system

PII Classification

- Identify sensitive fields (salary, SSN, health data)
- Define access control rules per role
- Decide what the LLM is allowed to see

Not All Knowledge Lives in Databases

The Other Half of Enterprise Knowledge

You have seen AI query **databases** — rows, columns, SQL. But think about where the bulk of your company's knowledge actually lives.

80%

of enterprise data is unstructured — policies, contracts, emails, reports

0%

of that is accessible through a dashboard

?

How many critical answers are trapped in PDFs nobody can search?

Demo: Document-Based Q&A

Prompt 1 — Policy Lookup

```
What is our policy on remote work for the Energy Division?
```

Demo Result: Policy Lookup with Citations

What the Agent Found

- **General policy** (Employee Handbook, Section 4.2, p. 12): Up to 2 days/week remote for eligible roles
- **Energy Division override** (Energy Division Policies, Section 2.1, p. 3): Field operations staff exempt; office staff limited to 1 day/week during Q4 operational reviews
- Agent synthesized both sources into a complete answer

Why This Matters

- Every claim has a citation — document, section, page number
- Cross-document reasoning: two sources, one answer
- A keyword search would likely find one source, not both
- The user can verify every claim against the original documents

Demo: Structured Data Extraction

Prompt 2 — Vendor Renewals

```
Which vendor contracts are up for renewal in the next 6 months?
What are the key terms and required notice periods?
```

Demo: Combining Documents + Database

Prompt 3 — Cross-System Synthesis

```
What's our contractual commitment to GE, and how does their
actual spending compare?
```

Demo Result: Documents + Database Combined

Capability	Source	What it provided
Document AI (RAG)	GE Master Service Agreement	$4.2M minimum commitment, discount tiers, shortfall clause, renewal terms
Database AI (SQL)	Salesforce + Legacy CRM + HubSpot	Actual spending by division and year: $4.28M (2024), $3.80M (2025)

> Neither source alone answers the question. The contract says what we **promised**. The database says what actually **happened**. The insight — a **$404K shortfall** with a renewal deadline **92 days away** — only emerges when both are combined.

How RAG Works

1

Chunk

Documents split into passages (paragraphs, sections, or clauses)

→

2

Embed

Each chunk converted to a numerical vector by an embedding model

→

3

Store

Vectors stored in a vector database for fast similarity search

→

4

Query

User's question is embedded and matched against stored chunks

→

5

Generate

Top matches sent to LLM as context; LLM answers with citations

RAG = Retrieval-Augmented Generation. The LLM does not memorize your documents — it **searches them at query time** and answers using only what it retrieves.

Breakout: How Scattered Is Your Data?

Breakout Discussion (9 minutes)

Map Your Landscape

- **3-5 core systems** (CRM, ERP, HR, finance, support)
- **2-3 document collections** that trap important knowledge (policies, contracts, procedures, board materials)
- Where is integration hardest?
- What cross-system questions would have the most business value?

Report Back

- **One common pattern** your group identified
- **One surprising idiosyncrasy** unique to someone's company
- Where does the most valuable knowledge live — databases or documents?

Bridge to Session 4

What Comes Next

Today: Asking Questions

- Single-system queries to databases
- Cross-system queries with fuzzy matching
- Document Q&A with citations
- Combined document + database queries

Session 4: Producing Deliverables

- Data → charts → narrative → board-ready memo
- The full reporting pipeline in a single prompt
- Automated weekly reports and executive summaries
- **The agent becomes your reporting pipeline**

You have seen AI **answer questions**. Next session: the same agent **produces complete deliverables** — memos, reports, risk assessments. The difference between a chat reply and a board presentation.

Homework

This Week

Homework for Session 4

1. **Log into the Meridian Corp app** and ask **10 business questions** — at least 3 cross-system and at least 2 from the document collection
2. **Write down** one question the agent answered well and one where it was unsatisfying or wrong — bring both to Session 4
3. **Complete your data map:** at least 5 systems your company uses, the top 3 cross-system questions you cannot answer today, and 2-3 document collections that trap important knowledge
4. *Optional viewing:* Demos 6-7 show building a RAG system from scratch and combining it with a database agent — deeper context for the document Q&A introduced today

Next session: We go from asking questions to producing **complete deliverables** — memos, reports, risk assessments. The agent becomes your reporting pipeline.

Session 3 Summary

AI Meets Your Data

30 sec

vs. 3 weeks for a cross-system question

80%

of enterprise data is unstructured — policies, contracts, documents

2 paths

Database AI + Document AI = the full picture